

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Monday 23 June 2025

Afternoon (Time: 1 hour 30 minutes) **Paper reference** **9FM0/4C**

Further Mathematics
Advanced
PAPER 4C: Further Mechanics 2

You must have:
 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

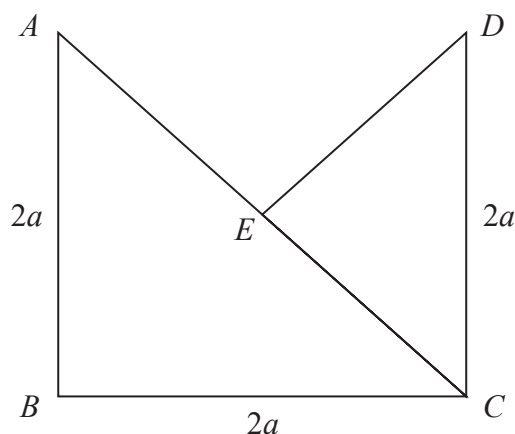


Figure 1

A uniform rod is cut into six pieces. The pieces are used to form the framework $ABCDE$ shown in Figure 1.

- All the pieces of the framework lie in the same plane.
- $AB = BC = CD = 2a$
- $AE = EC = ED = \sqrt{2}a$
- Point E is the midpoint of AC
- Angle $ABC = \text{angle } BCD = 90^\circ$

The distance of the centre of mass of the framework from AB is d

(a) Show that $d = \frac{12 + 7\sqrt{2}}{12 + 6\sqrt{2}}a$ (4)

The mass of the framework is M . A particle of mass kM is attached to the framework at B

The distance of the centre of mass of the loaded framework from AB is a

(b) Find the value of k (3)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



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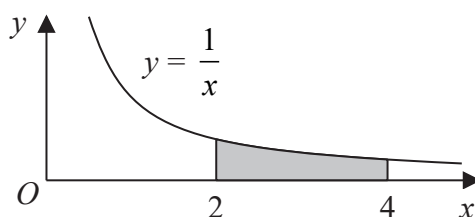


Figure 2

The shaded region shown in Figure 2 is bounded by the curve with equation $y = \frac{1}{x}$, the line with equation $x = 2$, the line with equation $x = 4$, and the x -axis. This region is rotated through 360° about the x -axis to form a solid.

This solid is used to model a uniform solid pedestal of height 2 m, upper radius 0.5 m and base radius 0.25 m.

Given that the volume of the pedestal is $\frac{\pi}{4} \text{ m}^3$

(a) show that the centre of mass of the pedestal is $(4 - 4 \ln 2)$ m from its base.

(4)

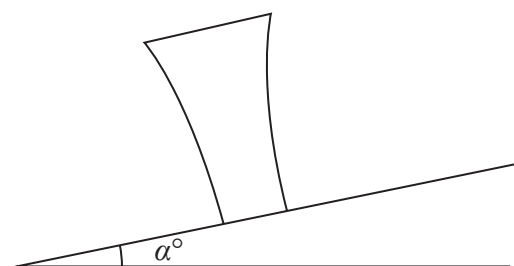


Figure 3

Diagram not drawn to scale

The pedestal is placed on a rough plane that is inclined at an angle α° to the horizontal. The plane base of the pedestal is in contact with the inclined plane, as shown in Figure 3.

The inclined plane is sufficiently rough to prevent the pedestal from sliding.

Given that the pedestal is on the point of toppling,

(b) find the value of α

(3)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



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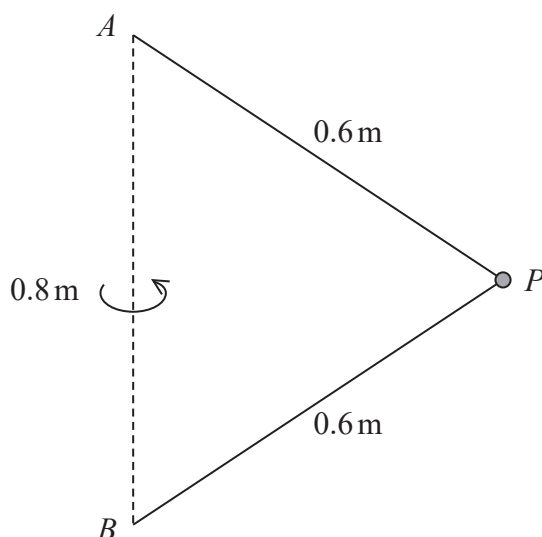


Figure 4

A particle P of mass 0.2 kg is attached to one end of a light inextensible string of length 0.6 m . The other end of the string is attached to the fixed point A .

A second light inextensible string of length 0.6 m has one end attached to P and the other end attached to the fixed point B .

The point B is vertically below A such that $AB = 0.8 \text{ m}$, as shown in Figure 4.

The particle moves in a horizontal circle with constant angular speed ω radians per second, with both strings taut.

Given that the tension in the string from P to A is twice the tension in the string from P to B , find the value of ω .

(7)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 7 marks)



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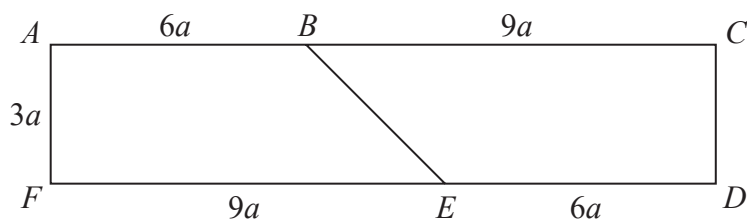


Figure 5

The uniform rectangular lamina $ACDF$ shown in Figure 5 has $AC = FD = 15a$ and $AF = CD = 3a$. The point B on AC is such that $AB = 6a$. The point E on FD is such that $ED = 6a$.

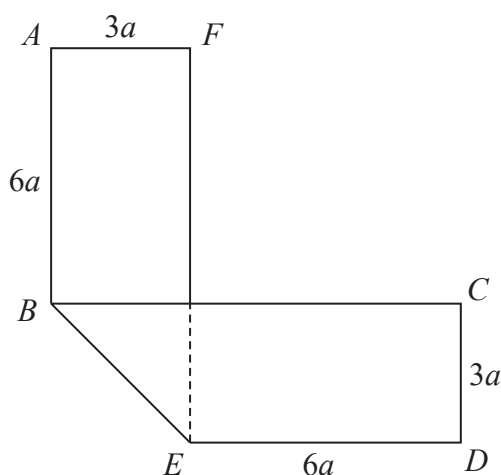


Figure 6

The rectangular lamina is folded along BE to form the folded lamina shown in Figure 6. The folded lamina has AB perpendicular to BC and the two sections, $ABEF$ and $BEDC$, of the lamina lie in the same plane.

(a) Show that the distance of the centre of mass of the folded lamina from EF is $\frac{2}{5}a$. (5)

(b) Explain why the centre of mass of the folded lamina lies on the perpendicular bisector of BE . (1)

The folded lamina is freely suspended from F and hangs in equilibrium.

(c) Find the size of the angle between FE and the downward vertical. (3)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 9 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 10 marks)



6. A uniform semicircular lamina has radius r and the midpoint of the diameter of the lamina is at O . The distance of the centre of mass of the lamina from O is d .

(a) Use algebraic integration to show that $d = \frac{4r}{3\pi}$ (4)

The uniform plane template T , shown shaded in Figure 7, is formed from a rectangle and a semicircular ring.

The rectangle has $AD = BC = 2a$ and $AB = DC = 8a$.

The midpoint of CD is O .

The semicircular ring has centre O , outer radius $OD = 4a$ and inner radius $OE = 2a$.

The template is modelled as a uniform lamina.

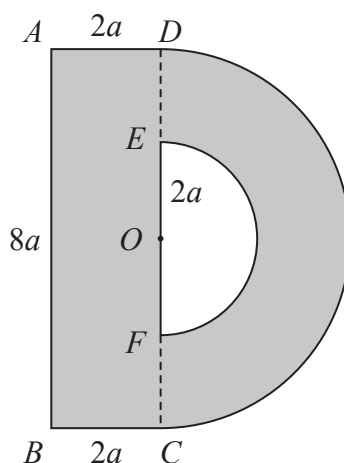


Figure 7

(b) Show that the distance of the centre of mass of T from DC is $\frac{32a}{3(8 + 3\pi)}$ (5)

The template is freely suspended from A . The weight of T is W newtons.

A horizontal force is applied to T at B so that T is held in equilibrium with AB vertical.

The force acts in the same vertical plane as T and has magnitude λW newtons.

(c) Find the value of λ (3)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 12 marks)



7. A particle P of mass 0.4 kg is moving in a straight line with simple harmonic motion. The maximum speed of P is 5 m s^{-1} and the maximum magnitude of acceleration of P is 12.5 m s^{-2} . The time taken for one complete oscillation is T seconds.

(a) Show that $T = \frac{4\pi}{5}$ (4)

- (b) Find the time within each complete oscillation for which P is more than 0.5 m from the centre of the oscillation. (4)

- (c) Find the kinetic energy of P at the instant when P is 0.2 m from the centre of the oscillation. (3)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 11 marks)



8.

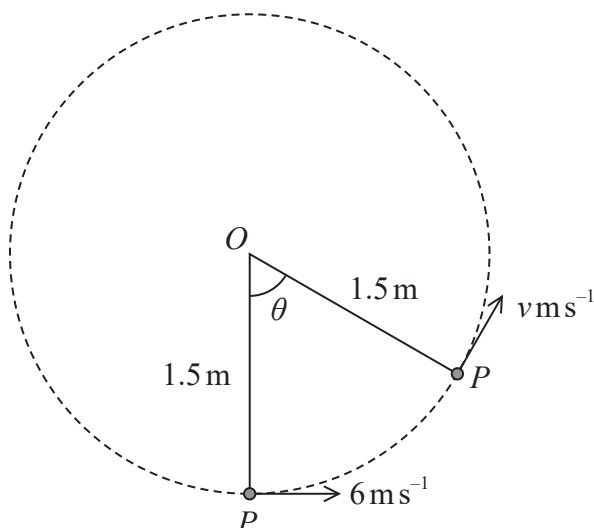
In this question use $g = 10 \text{ m s}^{-2}$ 

Figure 8

A particle P of mass 0.4 kg is attached to one end of a light inextensible string of length 1.5 m . The other end of the string is attached to a fixed point O . The particle is at rest vertically below O with the string taut.

The particle is then projected horizontally with speed 6 m s^{-1}

When the string has turned through an angle θ , the string is still taut and the speed of P is $v \text{ m s}^{-1}$, as shown in Figure 8.

(a) Show that when $v = 3\sqrt{2}$, the tension in the string is 6.4 N . (7)

(b) Find the magnitude of the acceleration of P when $v = 3\sqrt{2}$ (5)

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Question 8 continued

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Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



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Pearson Edexcel Level 3 GCE

Monday 24 June 2024

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Advanced
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Turn over ►

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1. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

A particle P moves along a straight line.
Initially P is at rest at the point O on the line.

At time t seconds, where $t \geq 0$

- the displacement of P from O is x metres
- the velocity of P is $v \text{ ms}^{-1}$ in the positive x direction
- the acceleration of P is $\frac{96}{(3t + 5)^3} \text{ ms}^{-2}$ in the positive x direction

(a) Show that, at time t seconds, $v = p - \frac{q}{(3t + 5)^2}$, where p and q are constants to be determined.

(4)

(b) Find the limiting value of v as t increases.

(1)

(c) Find the value of x when $t = 2$

(4)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 9 marks)



2.

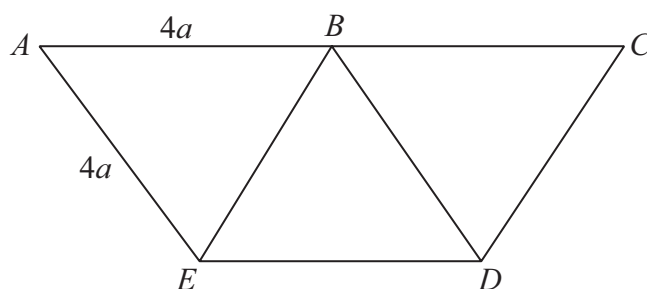


Figure 1

A uniform rod of length $28a$ is cut into seven identical rods each of length $4a$. These rods are joined together to form the rigid **framework** $ABCDEA$ shown in Figure 1.

All seven rods lie in the same plane.

The distance of the centre of mass of the framework from ED is d .

- (a) Show that $d = \frac{8\sqrt{3}}{7}a$ (4)

The weight of the framework is W .

The framework is freely pivoted about a horizontal axis through C .

The framework is held in equilibrium in a vertical plane, with AC vertical and A below C , by a **horizontal force** that is applied to the framework at A .

The force acts in the same vertical plane as the framework and has magnitude F .

- (b) Find F in terms of W . (3)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 7 marks)



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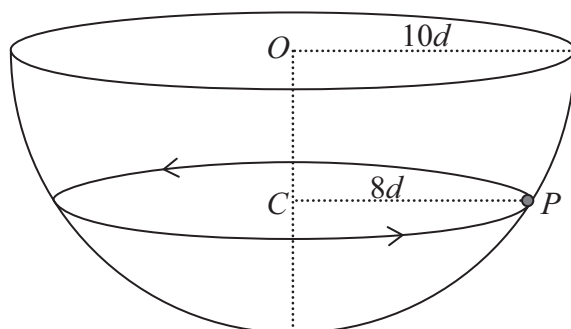


Figure 2

Figure 2 shows a hemispherical bowl of internal radius $10d$ that is fixed with its circular rim horizontal.

The centre of the circular rim is at the point O .

A particle P moves with **constant** angular speed on the smooth inner surface of the bowl.

The particle P moves in a horizontal circle with radius $8d$ and centre C .

(a) Find, in terms of g , the exact magnitude of the acceleration of P .

(6)

The time for P to complete one revolution is T .

(b) Find T in terms of d and g .

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 9 marks)



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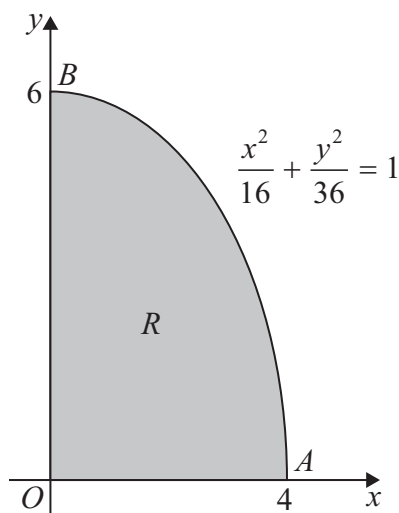


Figure 3

A uniform lamina OAB is in the shape of the region R .

Region R lies in the first quadrant and is bounded by the curve with equation

$\frac{x^2}{16} + \frac{y^2}{36} = 1$, the x -axis, and the y -axis, as shown shaded in Figure 3.

The point A is the point of intersection of the curve and the x -axis.

The point B is the point of intersection of the curve and the y -axis.

One unit on each axis represents 1 m.

The area of R is 6π

The centre of mass of R lies at the point with coordinates $(\bar{x}, \bar{y})$

(a) Use algebraic integration to show that $\bar{x} = \frac{16}{3\pi}$ (5)

(b) Use algebraic integration to find the exact value of $\bar{y}$ (4)

The lamina is freely suspended from A and hangs in equilibrium with OA at angle θ° to the downward vertical.

(c) Find the value of θ (3)

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 12 marks)



- (c) Find the value of S . (4)

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 11 marks)



6. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.



Figure 4

The shaded region, shown in Figure 4, is bounded by the x -axis, the line with equation $x = 6$, the line with equation $y = 2$ and the y -axis.

This region is rotated through 360° about the x -axis to form a solid of revolution. This solid is used to model a **non-uniform** cylinder of height 6 cm and radius 2 cm.

The mass per unit volume of the cylinder at the point (x, y, z) is $\lambda(x + 2) \text{ kg cm}^{-3}$, where $0 \leq x \leq 6$ and λ is a constant.

- (a) Show that the mass of the cylinder is $120\lambda\pi \text{ kg}$. (3)
- (b) Show that the centre of mass of the cylinder is 3.6 cm from O . (4)

The point O is the centre of one end of the cylinder. The point A is the centre of the other end of the cylinder.

A uniform solid hemisphere of radius 3 cm has density $\lambda \text{ kg cm}^{-3}$. The hemisphere is attached to the cylinder with the centre of its circular face in contact with the point A on the cylinder to form the model shown in Figure 5.

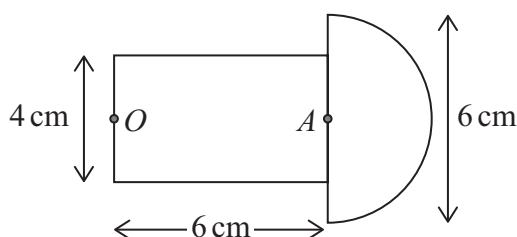


Figure 5

The model is placed with the end containing O on a rough inclined plane which is inclined at angle α° to the horizontal. The plane is sufficiently rough to prevent the model from sliding. The model is on the point of toppling.

- (c) Find the value of α . (6)

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Question 6 continued

Lined area for writing the answer to Question 6.

Question 6 continued

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Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 13 marks)



7.

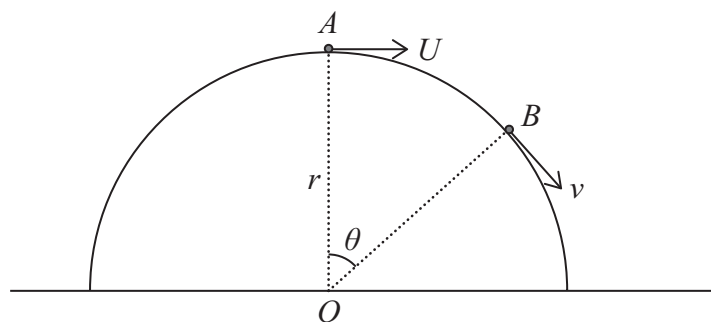


Figure 6

A smooth solid hemisphere has radius r and the centre of its plane face is O . The hemisphere is fixed with its plane face in contact with horizontal ground, as shown in Figure 6.

A small stone is at the point A , the highest point on the surface of the hemisphere.

The stone is projected horizontally from A with speed U .

The stone is still in contact with the hemisphere at the point B , where OB makes an angle θ with the upward vertical.

The speed of the stone at the instant it reaches B is v .

The stone is modelled as a particle P and air resistance is modelled as being negligible.

- (a) Use the model to find v^2 in terms of U , r , g and θ (3)

When P leaves the surface of the hemisphere, the speed of P is W .

Given that $U = \sqrt{\frac{2rg}{3}}$

- (b) show that $W^2 = \frac{8}{9}rg$ (5)

After leaving the surface of the hemisphere, P moves freely under gravity until it hits the ground.

- (c) Find the speed of P as it hits the ground, giving your answer in terms of r and g . (3)

At the instant when P hits the ground it is travelling at α° to the horizontal.

- (d) Find the value of α . (3)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS



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Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Monday 26th June 2023

Afternoon (Time: 1 hour 30 minutes)

Paper reference **9FM0/4C**

Further Mathematics

Advanced

PAPER 4C: Further Mechanics 2

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Turn over ►

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1. Three particles of masses $3m$, $4m$ and km are positioned at the points with coordinates $(2a, 3a)$, $(a, 5a)$ and $(2\mu a, \mu a)$ respectively, where k and μ are constants.

The centre of mass of the three particles is at the point with coordinates $(2a, 4a)$.

Find (i) the value of k

(ii) the value of μ

(6)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. A particle of mass 2 kg is moving in a straight line on a smooth horizontal surface under the action of a horizontal force of magnitude F newtons.

At time t seconds ($t > 0$),

- the particle is moving with speed $v \text{ ms}^{-1}$
- $F = 2 + v$

The time taken for the speed of the particle to increase from 5 ms^{-1} to 10 ms^{-1} is T seconds.

- (a) Show that $T = 2 \ln \frac{12}{7}$ (4)

The distance moved by the particle as its speed increases from 5 ms^{-1} to 10 ms^{-1} is D metres.

- (b) Find the exact value of D . (4)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



3. [In this question you may quote, without proof, the formula for the distance of the centre of mass of a uniform circular arc from its centre.]

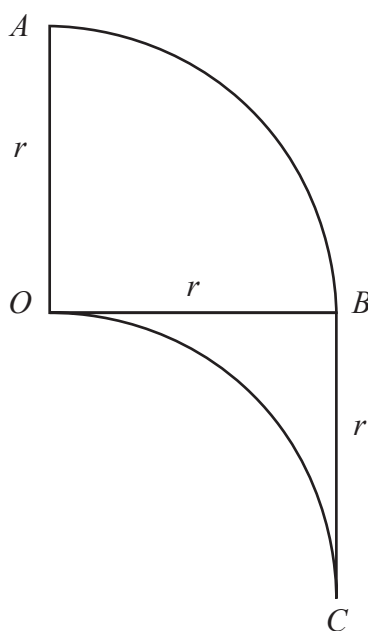


Figure 1

Five pieces of a uniform wire are joined together to form the rigid **framework** $OABCO$ shown in Figure 1, where

- OA , OB and BC are straight, with $OA = OB = BC = r$
- arc AB is one quarter of a circle with centre O and radius r
- arc OC is one quarter of a circle of radius r
- all five pieces of wire lie in the same plane

- (a) Show that the centre of mass of arc AB is a distance $\frac{2r}{\pi}$ from OA . (2)

Given that the distance of the centre of mass of the framework from OA is d ,

- (b) show that $d = \frac{7r}{2(3 + \pi)}$ (4)

The framework is freely pivoted at A .

The framework is held in equilibrium, with AO vertical, by a horizontal force of magnitude F which is applied to the framework at C .

Given that the weight of the framework is W

- (c) find F in terms of W (3)

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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4.

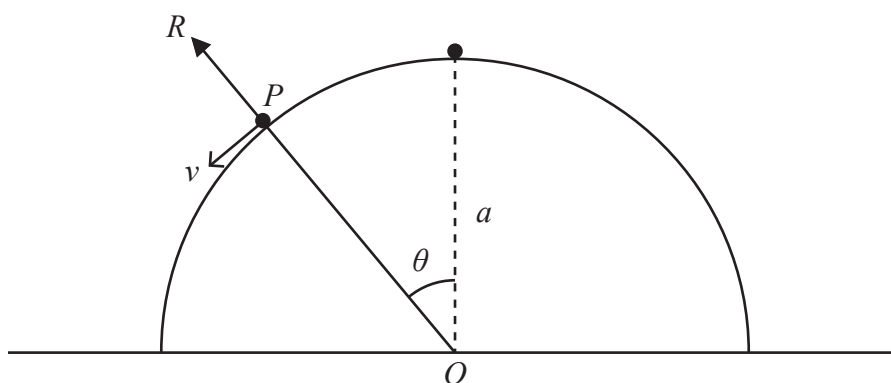


Figure 2

A smooth hemisphere of radius a is fixed on a horizontal surface with its plane face in contact with the surface. The centre of the plane face of the hemisphere is O .

A particle P of mass M is disturbed from rest at the highest point of the hemisphere.

When P is still on the surface of the hemisphere and the radius from O to P is at an angle θ to the vertical,

- the speed of P is v
- the normal reaction between the hemisphere and the particle is R , as shown in Figure 2.

(a) Show that $R = Mg(3\cos\theta - 2)$

(6)

(b) Find, in terms of a and g , the speed of the particle at the instant when the particle leaves the surface of the hemisphere.

(3)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 9 marks)



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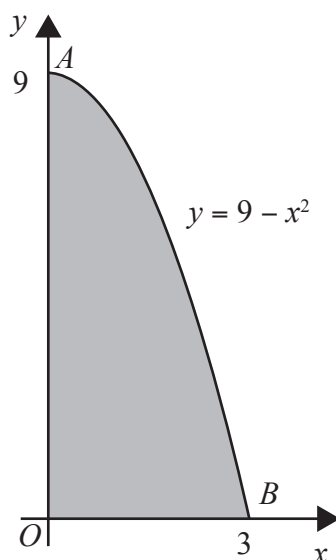


Figure 3

A uniform lamina OAB is modelled by the finite region bounded by the x -axis, the y -axis and the curve with equation $y = 9 - x^2$, for $x \geq 0$, as shown shaded in Figure 3. The unit of length on both axes is 1 m.

The area of the lamina is 18 m^2

- (a) Show that the centre of mass of the lamina is 3.6 m from OB .

[Solutions relying on calculator technology are not acceptable.]

(4)

A light string has one end attached to the lamina at O and the other end attached to the ceiling. A second light string has one end attached to the lamina at A and the other end attached to the ceiling.

The lamina hangs in equilibrium with the strings vertical and OA horizontal.

The weight of the lamina is W

The tension in the string attached to the lamina at A is λW

- (b) Find the value of λ

(3)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 7 marks)



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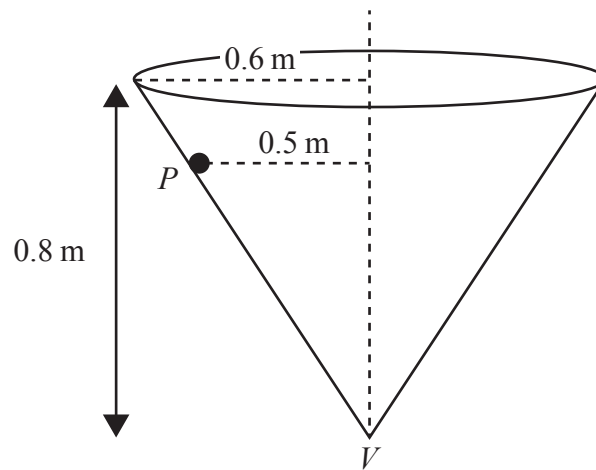


Figure 4

A hollow right circular cone, of internal base radius 0.6 m and height 0.8 m , is fixed with its axis vertical and its vertex V pointing downwards, as shown in Figure 4.

A particle P of mass $m\text{ kg}$ moves in a horizontal circle of radius 0.5 m on the rough inner surface of the cone.

The particle P moves with constant angular speed $\omega\text{ rad s}^{-1}$

The coefficient of friction between the particle P and the inner surface of the cone is 0.25

Find the greatest possible value of ω

(9)

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 9 marks)



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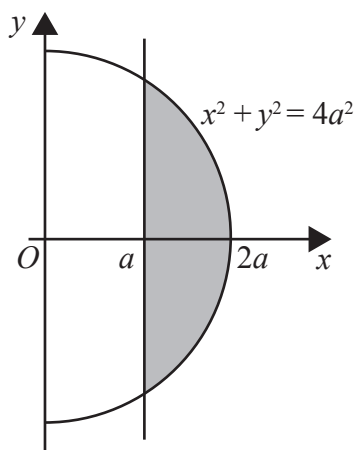


Figure 5

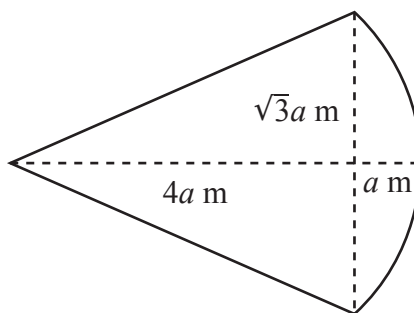


Figure 6

The shaded region shown in Figure 5 is bounded by the line with equation $x = a$ and the curve with equation $x^2 + y^2 = 4a^2$

This shaded region is rotated through 180° about the x -axis to form a solid of revolution.

This solid is used to model a dome with height a metres and base radius $\sqrt{3}a$ metres.

The dome is modelled as being non-uniform with the mass per unit volume of the dome at the point (x, y, z) equal to $\frac{\lambda}{x^2} \text{ kg m}^{-3}$, where $a \leq x \leq 2a$ and λ is a constant.

- (a) Show that the distance of the centre of mass of the dome from the centre of its plane face is $\left(4\ln 2 - \frac{5}{2}\right)a$ metres.

(6)

A solid uniform right circular cone has base radius $\sqrt{3}a$ metres and perpendicular height $4a$ metres. A toy is formed by attaching the plane surface of the dome to the plane surface of the cone, as shown in Figure 6.

The weight of the cone is kW and the weight of the dome is $2W$

The centre of mass of the toy is a distance d metres from the plane face of the dome.

- (b) Show that $d = \frac{|k + 5 - 8\ln 2|}{2 + k}a$

(4)

The toy is suspended from a point on the circumference of the plane face of the dome and hangs freely in equilibrium with the plane face of the dome at an angle α to the downward vertical.

Given that $\tan \alpha = \frac{1}{2\sqrt{3}}$

- (c) find the exact value of k .

(3)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 13 marks)



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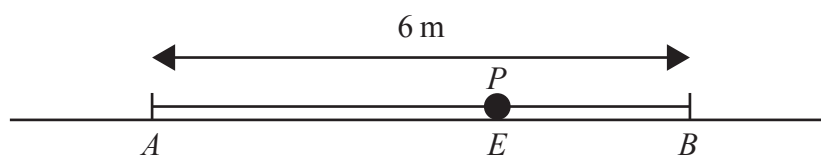


Figure 7

The fixed points A and B lie on a smooth horizontal surface with $AB = 6$ m.

A particle P has mass 0.3 kg.

One end of a light elastic string, of natural length 2 m and modulus of elasticity 20 N, is attached to P , and the other end is attached to A .

One end of another light elastic string, of natural length 2 m and modulus of elasticity 40 N, is attached to P and the other end is attached to B .

The particle P is at rest in equilibrium at the point E on the surface, as shown in Figure 7.

- (a) Show that $EB = \frac{8}{3}$ m. (3)

The particle P is now held at the midpoint of AB and released from rest.

- (b) Show that P oscillates with simple harmonic motion about the point E . (4)

The time between the instant when P is released and the instant when it first returns to the point E is S seconds.

- (c) Find the exact value of S . (3)

- (d) Find the length of time during one oscillation for which the speed of P is more than 2 m s⁻¹. (4)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 14 marks)

(TOTAL FOR PAPER IS 75 MARKS)



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Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper
reference

9FM0/4C

Further Mathematics

Advanced

PAPER 4C: Further Mechanics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Three particles of masses $2m$, $3m$ and km are placed at the points with coordinates $(3a, 2a)$, $(a, -4a)$ and $(-3a, 4a)$ respectively.

The centre of mass of the three particles lies at the point with coordinates $(\bar{x}, \bar{y})$.

(a) (i) Find $\bar{x}$ in terms of a and k

(ii) Find $\bar{y}$ in terms of a and k

(4)

Given that the distance of the centre of mass of the three particles from the point $(0, 0)$

is $\frac{1}{3}a$

(b) find the possible values of k

(2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. A cyclist and her cycle have a combined mass of 60 kg. The cyclist is moving along a straight horizontal road and is working at a constant rate of 200 W.

When she has travelled a distance x metres, her speed is $v \text{ ms}^{-1}$ and the magnitude of the resistance to motion is $3v^2 \text{ N}$.

(a) Show that $\frac{dv}{dx} = \frac{200 - 3v^3}{60v^2}$ (4)

The distance travelled by the cyclist as her speed increases from 2 ms^{-1} to 4 ms^{-1} is D metres.

(b) Find the exact value of D (3)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



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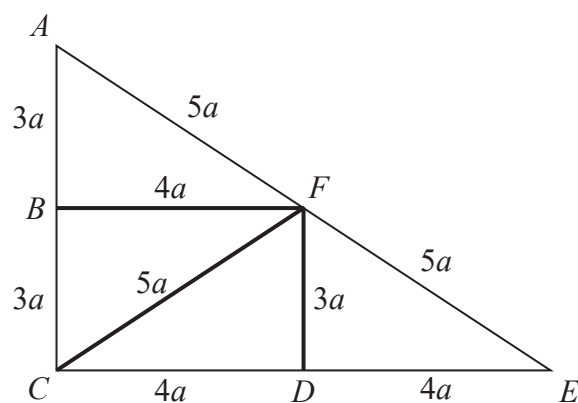


Figure 1

Nine uniform rods are joined together to form the rigid framework $ABCDEFA$, with $AB = BC = DF = 3a$, $BF = CD = DE = 4a$ and $AF = FE = CF = 5a$, as shown in Figure 1. All nine rods lie in the same plane.

The mass per unit length of each of the rods BF , CF and DF is twice the mass per unit length of each of the other six rods.

(a) Find the distance of the centre of mass of the framework from AC

(4)

The mass of the framework is M . A particle of mass kM is attached to the framework at E to form a loaded framework.

When the loaded framework is freely suspended from F , it hangs in equilibrium with CE horizontal.

(b) Find the exact value of k

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 7 marks)



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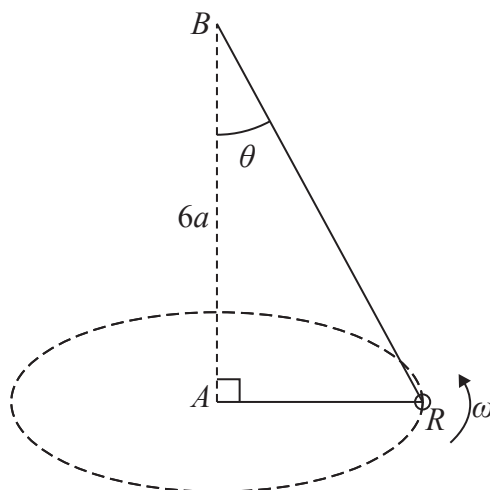


Figure 2

A small smooth ring R of mass m is threaded onto a light inextensible string. One end of the string is attached to a fixed point A and the other end of the string is attached to the fixed point B such that B is vertically above A and $AB = 6a$

The ring moves with constant angular speed ω in a horizontal circle with centre A . The string is taut and BR makes a constant angle θ with the downward vertical, as shown in Figure 2.

The ring is modelled as a particle.

Given that $\tan \theta = \frac{8}{15}$

(a) find, in terms of m and g , the magnitude of the tension in the string, (3)

(b) find ω in terms of a and g (5)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 8 marks)



5.

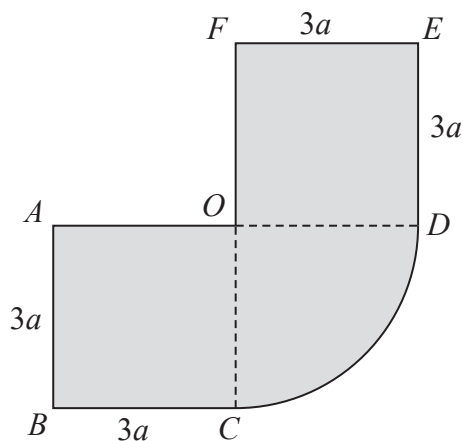


Figure 3

The uniform plane lamina shown in Figure 3 is formed from two squares, $ABCO$ and $ODEF$, and a sector ODC of a circle with centre O . Both squares have sides of length $3a$ and AO is perpendicular to OF . The radius of the sector is $3a$

[In part (a) you may use, without proof, any of the centre of mass formulae given in the formulae booklet.]

(a) Show that the distance of the centre of mass of the sector ODC from OC is $\frac{4a}{\pi}$ (3)

(b) Find the distance of the centre of mass of the lamina from FC (4)

The lamina is freely suspended from F and hangs in equilibrium with FC at an angle θ° to the downward vertical.

(c) Find the value of θ (4)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 11 marks)



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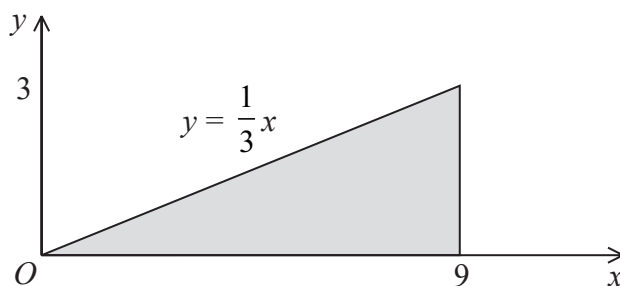


Figure 4

The shaded region shown in Figure 4 is bounded by the x -axis, the line with equation $x = 9$ and the line with equation $y = \frac{1}{3}x$. This shaded region is rotated through 360° about the x -axis to form a solid of revolution. This solid of revolution is used to model a solid right circular cone of height 9 cm and base radius 3 cm.

The cone is non-uniform and the mass per unit volume of the cone at the point (x, y, z) is $\lambda x \text{ kg cm}^{-3}$, where $0 \leq x \leq 9$ and λ is constant.

(a) Find the distance of the centre of mass of the cone from its vertex.

(6)

A toy is made by joining the circular plane face of the cone to the circular plane face of a uniform solid hemisphere of radius 3 cm, so that the centres of the two plane surfaces coincide.

The weight of the cone is W newtons and the weight of the hemisphere is kW newtons.

When the toy is placed on a smooth horizontal plane with any point of the curved surface of the hemisphere in contact with the plane, the toy will remain at rest.

(b) Find the value of k

(4)



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Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 10 marks)



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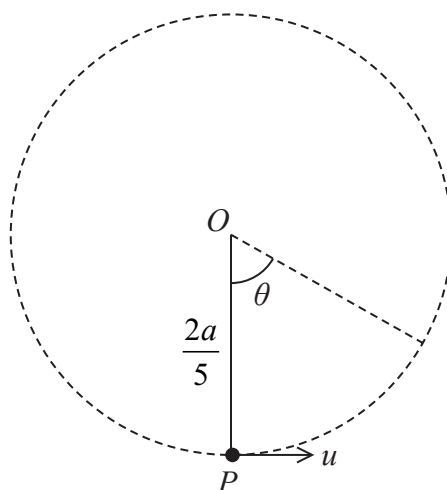


Figure 5

A package P of mass m is attached to one end of a string of length $\frac{2a}{5}$. The other end of the string is attached to a fixed point O . The package hangs at rest vertically below O with the string taut and is then projected horizontally with speed u , as shown in Figure 5.

When OP has turned through an angle θ and the string is still taut, the tension in the string is T .

The package is modelled as a particle and the string as being light and inextensible.

(a) Show that $T = 3mg \cos \theta - 2mg + \frac{5mu^2}{2a}$ (6)

Given that P moves in a complete vertical circle with centre O

(b) find, in terms of a and g , the minimum possible value of u (2)

Given that $u = 2\sqrt{ag}$

(c) find, in terms of g , the magnitude of the acceleration of P at the instant when OP is horizontal. (3)

(d) Apart from including air resistance, suggest one way in which the model could be refined to make it more realistic. (1)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 12 marks)



8.

Throughout this question, use $g = 10 \text{ m s}^{-2}$

A light elastic string has natural length 1.25 m and modulus of elasticity 25 N.

A particle P of mass 0.5 kg is attached to one end of the string. The other end of the string is attached to a fixed point A . Particle P hangs freely in equilibrium with P vertically below A .

The particle is then pulled vertically down to a point B and released from rest.

- (a) Show that, while the string is taut, P moves with simple harmonic motion with period $\frac{\pi}{\sqrt{10}}$ seconds.

The maximum kinetic energy of P during the subsequent motion is 2.5 J.

- (b) Show that $AB = 2\text{ m}$ (3)

The particle returns to B for the first time T seconds after it was released from rest at B

- (c) Find the value of T (5)

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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS



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Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Time 1 hour 30 minutes	Paper reference	9FM0/4C	
Further Mathematics Advanced PAPER 4C: Further Mechanics 2			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

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1.

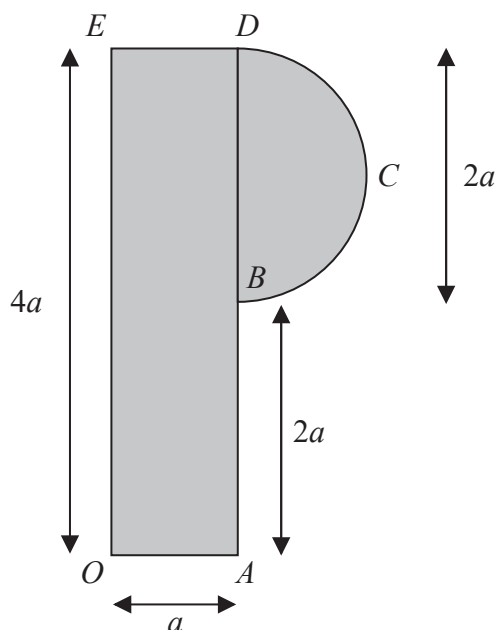


Figure 1

A letter P from a shop sign is modelled as a uniform plane lamina which consists of a rectangular lamina, $OABDE$, joined to a semicircular lamina, BCD , along its diameter BD .

$OA = ED = a$, $AB = 2a$, $OE = 4a$, and the diameter $BD = 2a$, as shown in Figure 1.

Using the model,

(a) find, in terms of π and a , the distance of the centre of mass of the letter P,

from (i) OE

(ii) OA

(6)

The letter P is freely suspended from O and hangs in equilibrium. The angle between OE and the downward vertical is α .

Using the model,

(b) find the exact value of $\tan \alpha$

(2)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 8 marks)



2. At time $t = 0$, a small stone P of mass m is released from rest and falls vertically through the air. At time t , the speed of P is v and the resistance to the motion of P from the air is modelled as a force of magnitude kv^2 , where k is a constant.

(a) Show that $t = \frac{V}{2g} \ln\left(\frac{V+v}{V-v}\right)$ where $V^2 = \frac{mg}{k}$ (4)

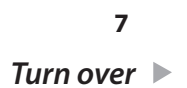
- (b) Give an interpretation of the value of V , justifying your answer. (2)

At time t , P has fallen a distance s .

(c) Show that $s = \frac{V^2}{2g} \ln\left(\frac{V^2}{V^2 - v^2}\right)$ (4)



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Question 2 continued

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Question 2 continued

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(Total for Question 2 is 10 marks)



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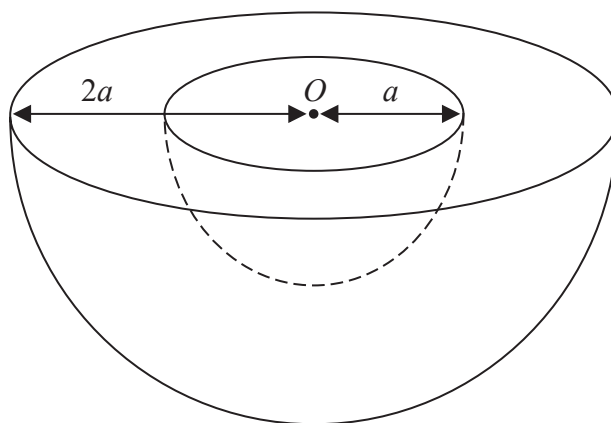


Figure 2

A uniform solid hemisphere H has radius $2a$. A solid hemisphere of radius a is removed from the hemisphere H to form a bowl. The plane faces of the hemispheres coincide and the centres of the two hemispheres coincide at the point O , as shown in Figure 2.

The centre of mass of the bowl is at the point G .

(a) Show that $OG = \frac{45a}{56}$

(4)

Figure 3 below shows a cross-section of the bowl which is resting in equilibrium with a point P on its curved surface in contact with a rough plane. The plane is inclined to the horizontal at an angle α and is sufficiently rough to prevent the bowl from slipping. The line OG is horizontal and the points O , G and P lie in a vertical plane which passes through a line of greatest slope of the inclined plane.

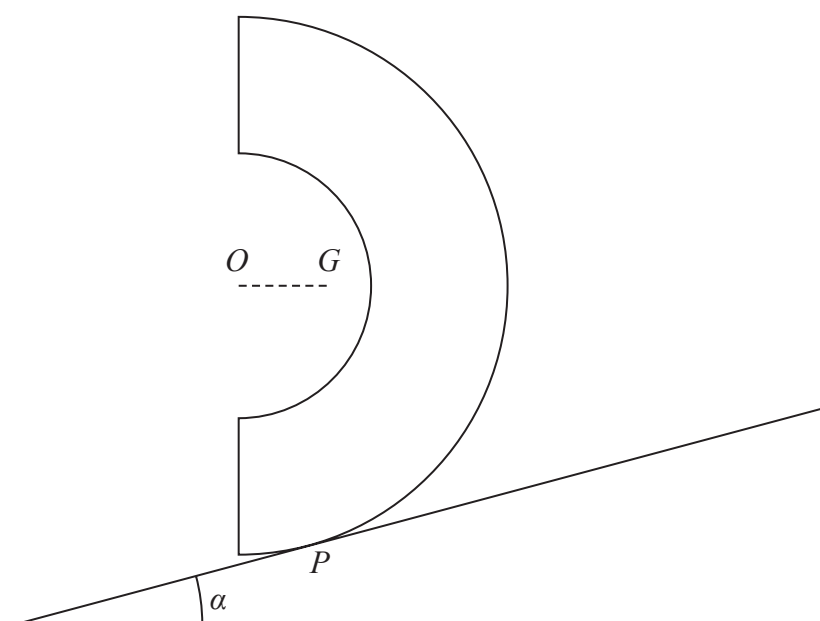


Figure 3

(b) Find the size of α , giving your answer in degrees to 3 significant figures.

(2)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 6 marks)



4.

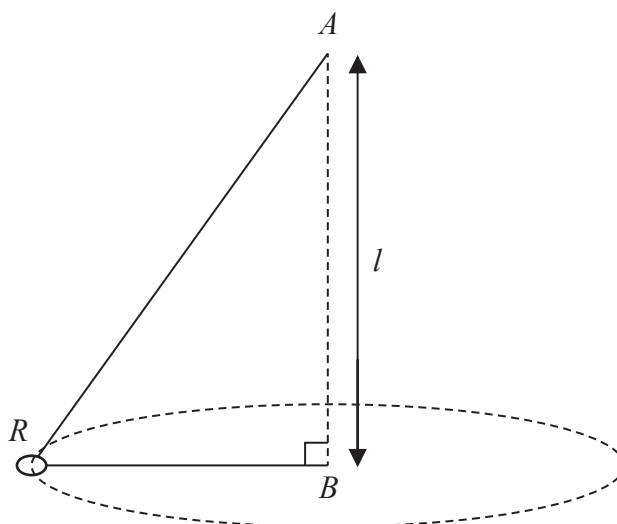


Figure 4

One end of a light inextensible string of length $2l$ is attached to a fixed point A . A small smooth ring R of mass m is threaded on the string and the other end of the string is attached to a fixed point B . The point B is vertically below A , with $AB = l$. The ring is then made to move with constant speed V in a horizontal circle with centre B . The string is taut and BR is horizontal, as shown in Figure 4.

- (a) Show that $BR = \frac{3l}{4}$
- (2)**

Given that air resistance is negligible,

- (b) find, in terms of m and g , the tension in the string, (4)
- (c) find V in terms of g and l . (4)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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5. A light inextensible string of length a has one end attached to a fixed point O . The other end of the string is attached to a small stone of mass m . The stone is held with the string taut and horizontal. The stone is then projected vertically upwards with speed U .

The stone is modelled as a particle and air resistance is modelled as being negligible.

Assuming that the string does not break, use the model to

- (a) find the least value of U so that the stone will move in complete vertical circles. (6)

The string will break if the tension in it is equal to $\frac{11mg}{2}$

Given that $U = 2\sqrt{ag}$, use the model to

- (b) find the total angle that the string has turned through, from when the stone is projected vertically upwards, to when the string breaks,

- (c) find the magnitude of the acceleration of the stone at the instant just before the string breaks. (4)



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Question 5 continued

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6. A light elastic string, of natural length l and modulus of elasticity $2mg$, has one end attached to a fixed point A and the other end attached to a particle P of mass m . The particle P hangs in equilibrium at the point O .

(a) Show that $AO = \frac{3l}{2}$

(2)

The particle P is pulled down vertically from O to the point B , where $OB = l$, and released from rest.

Air resistance is modelled as being negligible.

Using the model,

- (b) prove that P begins to move with simple harmonic motion about O with

$$\text{period } \pi \sqrt{\frac{2l}{g}} \quad (5)$$

The particle P first comes to instantaneous rest at the point C .

Using the model,

- (c) find the length BC in terms of l ,

- (d) find, in terms of l and g , the exact time it takes P to move directly from B to C .



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Question 6 continued

Handwriting practice area with horizontal lines.



Question 6 continued

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Question 6 continued

(Total for Question 6 is 16 marks)



7. [In this question, you may assume that the centre of mass of a circular arc, radius r , with angle at centre 2α , is a distance $\frac{r \sin \alpha}{\alpha}$ from the centre.]

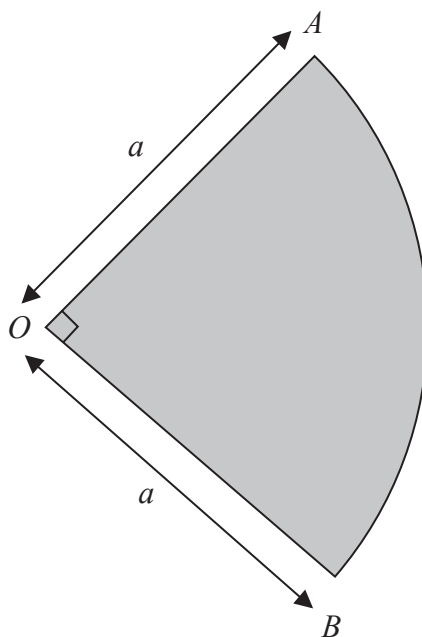


Figure 5

A thin non-uniform metal plate is in the shape of a sector OAB of a circle with centre O and radius a . The angle $AOB = \frac{\pi}{2}$, as shown in Figure 5.

The plate is modelled as a non-uniform lamina.

The mass per unit area of the lamina, at any point P of the lamina, is modelled as

$k(OP)^2$, where $k = \frac{4\lambda}{\pi a^4}$ and λ is a constant.

Using the model,

- (a) find the mass of the plate in terms of λ , (5)

- (b) find, in terms of a , the distance of the centre of mass of the plate from O . (4)



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Question 7 continued

Handwriting practice area with horizontal lines.



Question 7 continued

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Question 7 continued

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(Total for Question 7 is 9 marks)

TOTAL FOR PAPER IS 75 MARKS



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Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Friday 23 October 2020

Afternoon (Time: 1 hour 30 minutes)

Paper Reference **9FM0/4C**

Further Mathematics

Advanced

Paper 4C: Further Mechanics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
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Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
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Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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P 6 2 6 8 4 R A 0 1 2 8



Pearson

1. Three particles of masses $3m$, $4m$ and $2m$ are placed at the points $(-2, 2)$, $(3, 1)$ and (p, p) respectively.

The value of p is such that the distance of the centre of mass of the three particles from the point $(0, 0)$ is as small as possible.

Find the value of p .

(7)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2.

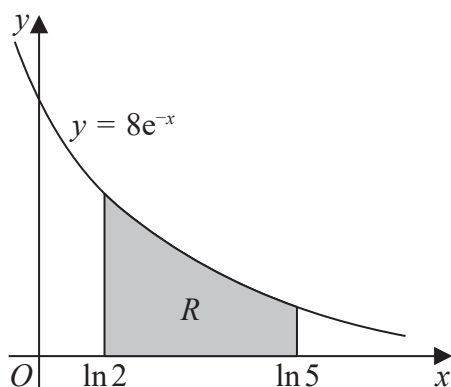


Figure 1

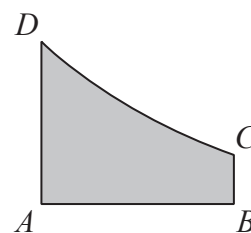


Figure 2

A uniform plane figure R , shown shaded in Figure 1, is bounded by the x -axis, the line with equation $x = \ln 5$, the curve with equation $y = 8e^{-x}$ and the line with equation $x = \ln 2$. The unit of length on each axis is one metre.

The area of R is 2.4 m^2

The centre of mass of R is at the point with coordinates $(\bar{x}, \bar{y})$.

(a) Use algebraic integration to show that $\bar{y} = 1.4$

(4)

Figure 2 shows a uniform lamina $ABCD$, which is the same size and shape as R . The lamina is freely suspended from C and hangs in equilibrium with CB at an angle θ° to the downward vertical.

(b) Find the value of θ

(6)

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Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 10 marks)



3. A particle P of mass 0.5 kg is moving along the positive x -axis in the direction of x increasing. At time t seconds ($t \geq 0$), P is x metres from the origin O and the speed of P is $v \text{ m s}^{-1}$. The resultant force acting on P is directed towards O and has magnitude $kv^2 \text{ N}$, where k is a positive constant.

When $x = 1$, $v = 4$ and when $x = 2$, $v = 2$

- (a) Show that $v = ab^x$, where a and b are constants to be found.

(6)

The time taken for the speed of P to decrease from 4 m s^{-1} to 2 m s^{-1} is T seconds.

- (b) Show that $T = \frac{1}{4 \ln 2}$

(4)



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Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 10 marks)



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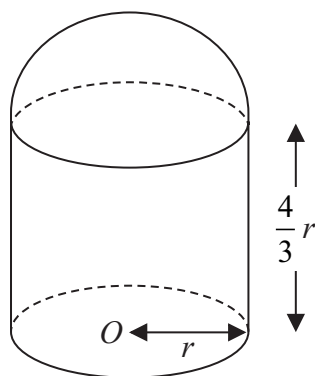


Figure 3

A uniform solid cylinder of base radius r and height $\frac{4}{3}r$ has the same density as a uniform solid hemisphere of radius r . The plane face of the hemisphere is joined to a plane face of the cylinder to form the composite solid S shown in Figure 3. The point O is the centre of the plane face of S .

- (a) Show that the distance from O to the centre of mass of S is $\frac{73}{72}r$ (4)

The solid S is placed with its plane face on a rough horizontal plane. The coefficient of friction between S and the plane is μ . A horizontal force P is applied to the highest point of S . The magnitude of P is gradually increased.

- (b) Find the range of values of μ for which S will slide before it starts to tilt. (5)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 9 marks)



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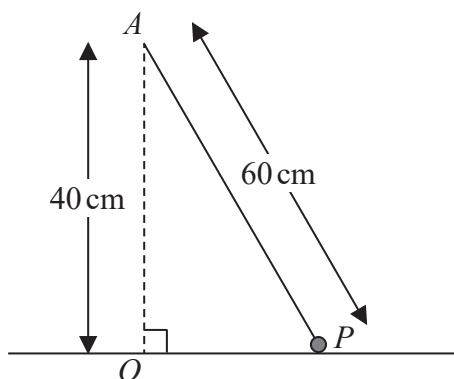


Figure 4

A particle P of mass 0.75 kg is attached to one end of a light inextensible string of length 60 cm . The other end of the string is attached to a fixed point A that is vertically above the point O on a smooth horizontal table, such that $OA = 40 \text{ cm}$. The particle remains in contact with the table, with the string taut, and moves in a horizontal circle with centre O , as shown in Figure 4.

The particle is moving with a constant angular speed of $3 \text{ radians per second}$.

- (a) Find
- (i) the tension in the string,
 - (ii) the normal reaction between P and the table.
- (7)

The angular speed of P is now gradually increased.

- (b) Find the angular speed of P at the instant P loses contact with the table.
- (4)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 11 marks)



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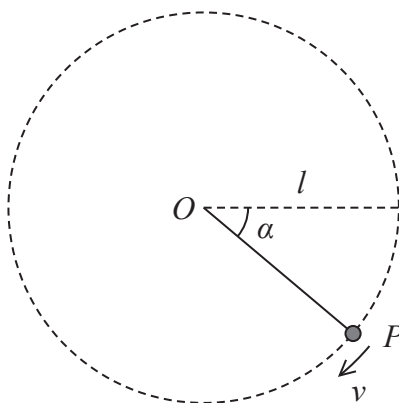


Figure 5

A particle P of mass m is attached to one end of a light inextensible string of length l . The other end of the string is attached to a fixed point O . The particle is held with the string taut and OP horizontal. The particle is then projected vertically downwards with speed u , where $u^2 = \frac{9}{5}gl$. When OP has turned through an angle α and the string is still taut, the speed of P is v , as shown in Figure 5. At this instant the tension in the string is T .

- (a) Show that $T = 3mg \sin \alpha + \frac{9}{5}mg$ (6)
- (b) Find, in terms of g and l , the speed of P at the instant when the string goes slack. (3)
- (c) Find, in terms of l , the greatest vertical height reached by P above the level of O . (4)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 13 marks)



7. A light elastic spring has natural length l and modulus of elasticity $4mg$. A particle P of mass m is attached to one end of the spring. The other end of the spring is attached to a fixed point A . The point B is vertically below A with $AB = \frac{7}{4}l$. The particle P is released from rest at B .

(a) Show that P moves with simple harmonic motion with period $\pi\sqrt{\frac{l}{g}}$ (7)

(b) Find, in terms of m , l and g , the maximum kinetic energy of P during the motion. (3)

(c) Find the time within each complete oscillation for which the length of the spring is less than l . (5)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

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(Total for Question 7 is 15 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Tuesday 25 June 2019			
Morning (Time: 1 hour 30 minutes)		Paper Reference 9FM0/4C	
Further Mathematics Advanced Paper 4C: Further Mechanics 2			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Information

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- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
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Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

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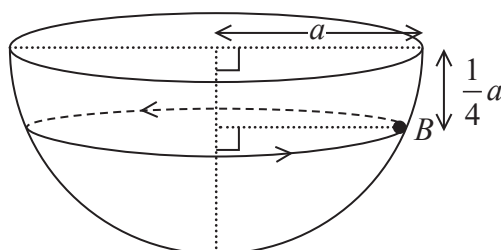


Figure 1

A hemispherical shell of radius a is fixed with its rim uppermost and horizontal. A small bead, B , is moving with constant angular speed, ω , in a horizontal circle on the smooth inner surface of the shell. The centre of the path of B is at a distance $\frac{1}{4}a$ vertically below the level of the rim of the hemisphere, as shown in Figure 1.

Find the magnitude of ω , giving your answer in terms of a and g .

(6)

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



2. A particle, P , of mass 0.4 kg is moving along the positive x -axis, in the positive x direction under the action of a single force. At time t seconds, $t > 0$, P is x metres from the origin O and the speed of P is $v \text{ m s}^{-1}$. The force is acting in the direction of x increasing and has magnitude $\frac{k}{v}$ newtons, where k is a constant.

At $x = 3$, $v = 2$ and at $x = 6$, $v = 2.5$

- (a) Show that $v^3 = \frac{61x + 9}{24}$ (6)

The time taken for the speed of P to increase from 2 m s^{-1} to 2.5 m s^{-1} is T seconds.

- (b) Use algebraic integration to show that $T = \frac{81}{61}$ (4)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 10 marks)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 11 marks)



4. A flagpole, AB , is 4 m long. The flagpole is modelled as a non-uniform rod so that, at a distance x metres from A , the mass per unit length of the flagpole, $m \text{ kg m}^{-1}$, is given by $m = 18 - 3x$.

(a) Show that the mass of the flagpole is 48 kg.

(3)

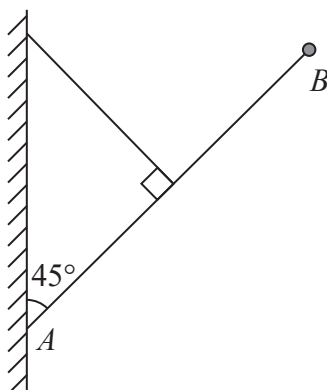


Figure 3

The end A of the flagpole is fixed to a point on a vertical wall. A cable has one end attached to the midpoint of the flagpole and the other end attached to a point on the wall that is vertically above A . The cable is perpendicular to the flagpole. The flagpole and the cable lie in the same vertical plane that is perpendicular to the wall. A small ball of mass 4 kg is attached to the flagpole at B . The cable holds the flagpole and ball in equilibrium, with the flagpole at 45° to the wall, as shown in Figure 3.

The tension in the cable is T newtons.

The cable is modelled as a light inextensible string and the ball is modelled as a particle.

(b) Using the model, find the value of T .

(8)

(c) Give a reason why the answer to part (b) is not likely to be the true value of T .

(1)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 12 marks)



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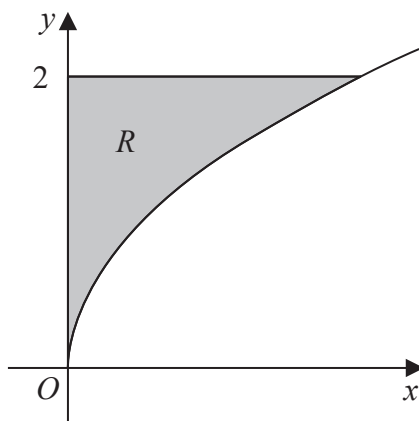


Figure 4

The region R , shown shaded in Figure 4, is bounded by part of the curve with equation $y^2 = 2x$, the line with equation $y = 2$ and the y -axis. The unit of length on both axes is one centimetre. A uniform solid, S , is formed by rotating R through 360° about the y -axis.

Given that the volume of S is $\frac{8}{5}\pi \text{ cm}^3$,

- (a) show that the centre of mass of S is $\frac{1}{3}$ cm from its plane face.

(4)

A uniform solid cylinder, C , has base radius 2 cm and height 4 cm. The cylinder C is attached to S so that the plane face of S coincides with a plane face of C , to form the paperweight P , shown in Figure 5. The density of the material used to make S is three times the density of the material used to make C .

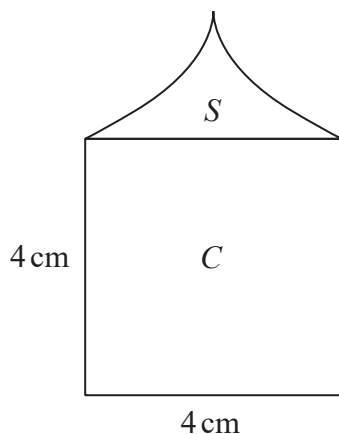


Figure 5

The plane face of P rests in equilibrium on a desk lid that is inclined at an angle θ° to the horizontal. The lid is sufficiently rough to prevent P from slipping. Given that P is on the point of toppling,

- (b) find the value of θ .

(7)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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[illegible]

6. The points A and B lie on a smooth horizontal surface with $AB = 4.5$ m.

A light elastic string has natural length 1.5 m and modulus of elasticity 15 N. One end of the string is attached to A and the other end of the string is attached to B . A particle, P , of mass 0.2 kg, is attached to the stretched string so that APB is a straight line and $AP = 1.5$ m. The particle rests in equilibrium on the surface.

The particle is now moved directly towards A and is held on the surface so APB is a straight line with $AP = 1$ m.

The particle is released from rest.

- (a) Prove that P moves with simple harmonic motion.

(5)

- (b) Find

- (i) the maximum speed of P during the motion,
(ii) the maximum acceleration of P during the motion.

(3)

- (c) Find the total time, in each complete oscillation of P , for which the speed of P is greater than 5 m s^{-1} .

(5)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 13 marks)



7. A particle, P , of mass m is attached to one end of a light rod of length L . The other end of the rod is attached to a fixed point O so that the rod is free to rotate in a vertical plane about O . The particle is held with the rod horizontal and is then projected vertically downwards with speed u . The particle first comes to instantaneous rest at the point A .

(a) Explain why the acceleration of P at A is perpendicular to OA .

(1)

At the instant when P is at the point A the acceleration of P is in a direction making an angle θ with the horizontal. Given that $u^2 = \frac{2gL}{3}$,

(b) find

(i) the magnitude of the acceleration of P at the point A ,

(ii) the size of θ .

(6)

(c) Find, in terms of m and g , the magnitude of the tension in the rod at the instant when P is at its lowest point.

(5)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Mock Paper Set 1			
(Time: 1 hour 30 minutes)		Paper Reference 9FM0/4C	
Further Mathematics Advanced Paper 4C: Further Mechanics 2			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
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Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

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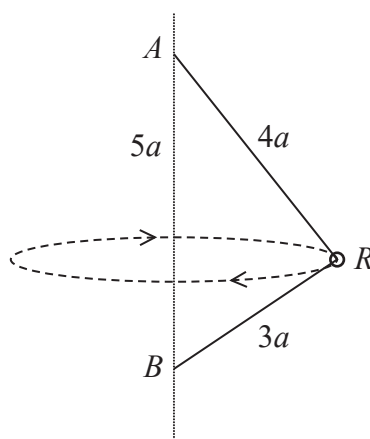


Figure 1

A small smooth ring, R , of mass m is threaded on a light inextensible string of length $7a$. The string has its ends attached to two fixed points, A and B , where A is vertically above B and $AB = 5a$. The ring moves in a horizontal circle about AB , with the string taut and $AR = 4a$, as shown in Figure 1.

The ring moves with constant angular speed ω .

Find the magnitude of ω , giving your answer in terms of a and g .

(6)

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2.

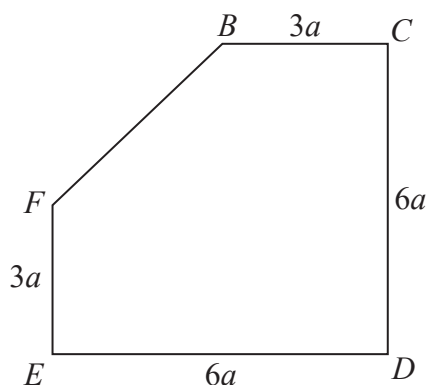


Figure 2

The uniform lamina, L , with $BC = EF = 3a$ and $CD = DE = 6a$, shown in Figure 2, is formed by removing an isosceles triangle, whose equal sides are of length $3a$, from a square of side $6a$.

- (a) Show that the centre of mass of L is $\frac{23}{7}a$ from EF . (4)

A uniform triangular lamina, ABF , made from a different material to L , is fixed to L to form the square template, $ABCDEF$, shown in Figure 3.

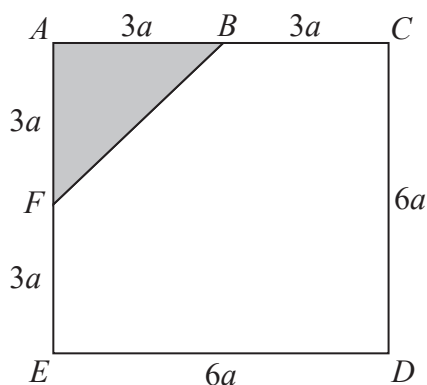


Figure 3

The lamina L has mass M per unit area. The triangular lamina ABF has mass $3M$ per unit area.

The template is freely suspended from C and hangs in equilibrium, with CD at an angle θ° to the downward vertical.

- (b) Find the value of θ . (6)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 10 marks)



3. A particle, P , moves in a straight line with simple harmonic motion about a fixed centre O . The maximum speed of P during the motion is 1.2 m s^{-1} . The maximum magnitude of the acceleration of P during the motion is 4.8 m s^{-2} .

Find

- (a) (i) the period of the motion, (5)
(ii) the amplitude of the motion.
- (b) the speed of P when P is 0.1 m from O (2)
- (c) the length of time within one oscillation for which P is less than 0.1 m from O (3)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 10 marks)



4. A tree trunk is 6 m long and has mass 630 kg.

The tree trunk is modelled as a non-uniform rod such that, at a distance x metres from the base of the tree trunk, the mass per unit length of the tree trunk, $\lambda \text{ kg m}^{-1}$, is given by the

formula $\lambda = k \left(2 - \frac{x}{12} \right)$, where k is a positive constant.

- (a) Show that $k = 60$

(3)

The tree trunk rests horizontally in equilibrium on two supports, each of which is 2 m from an end, as shown in Figure 4.

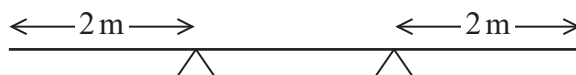


Figure 4

- (b) Using the model, find the magnitude of each of the reactions between the tree trunk and the supports.

(8)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 11 marks)



5. A particle, P , is moving along the positive x -axis. At time t seconds, $OP = x$ metres, the velocity of P is $v \text{ ms}^{-1}$ in the direction of x increasing, and the acceleration of P is

$$\left(\frac{10}{x^2} - \frac{4}{x^3}\right) \text{ ms}^{-2}, \text{ where } x \geq 1, \text{ in the direction of } x \text{ increasing.}$$

At time $t = 0$ seconds, $x = 1$ and $v = 3$

- (a) Show that $v = \frac{5x - 2}{x}$, justifying your answer carefully. (6)

At time T seconds, $x = 4$

- (b) Show that $T = \frac{3}{5} + \frac{2}{25} \ln 6$ (6)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 12 marks)



6.

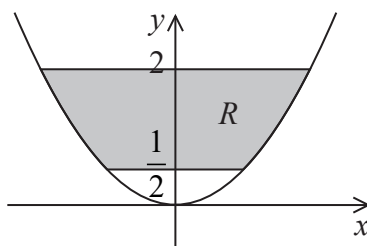


Figure 5

The region R , shown shaded in Figure 5, is bounded by the curve with equation $y = \frac{1}{2}x^2$, the line with equation $y = 2$ and the line with equation $y = \frac{1}{2}$. The unit of length on each axis is one centimetre.

A uniform solid, S , is formed by rotating R through 180° about the y -axis.

Given that the volume of S is $\frac{15\pi}{4} \text{ cm}^3$ and using algebraic integration,

(a) show that the centre of mass of S is exactly 1.4 cm from the x -axis.

(5)

A uniform solid hemisphere of radius 2 cm, with the same density as S , is joined to S to form the composite solid B . The plane face of the hemisphere coincides with the larger plane face of S .

The plane face of B rests on a rough inclined plane that is at an angle α° to the horizontal. The inclined plane is sufficiently rough to prevent B from slipping.

Given that B is on the point of toppling,

(b) find the value of α .

(7)



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Question 6 continued

Lined area for writing the answer to Question 6.



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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 12 marks)



7.

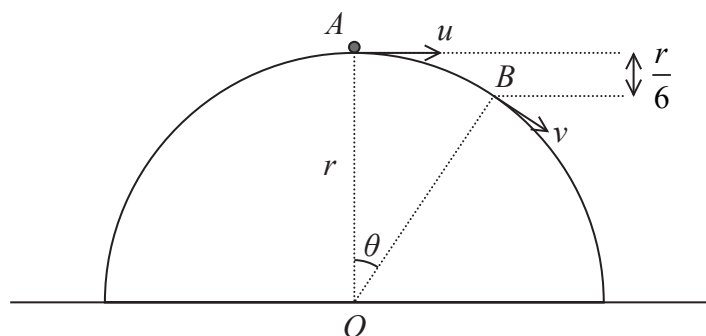


Figure 6

A smooth solid hemisphere, of radius r and centre O , is fixed with its plane face on a horizontal table. The highest point on the surface of the hemisphere is A .

A particle, P , of mass m is projected horizontally with speed u , where $u < \sqrt{gr}$, from A and moves along the surface of the hemisphere.

The particle leaves the surface of the hemisphere at the point B , which is a vertical distance $\frac{r}{6}$ below the level of A . The speed of P at the instant it reaches B is v , and the angle AOB is θ , as shown in Figure 6.

Assuming that air resistance may be ignored,

- (a) show that $v^2 = \frac{5gr}{6}$ (4)
- (b) explain the need for the restriction $u < \sqrt{gr}$ (1)
- (c) find u in terms of g and r (4)
- (d) find the direction of motion of P at the instant it hits the table. (5)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

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Level 3 GCE**

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Candidate Number

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Mock Paper Set 2

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/4C**

Further Mathematics

Advanced

Paper 4C: Further Mechanics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

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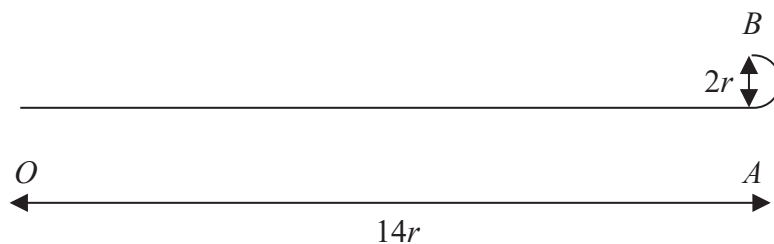


Figure 1

A hockey stick is modelled as a uniform rod OA of length $14r$ joined to a uniform semicircular arc AB of diameter $2r$, as shown in Figure 1. The rod and the arc lie in the same plane and are made of the same material.

- (a) Find, in terms of π and r , the distance of the centre of mass of the hockey stick from the line AB .

(5)

The hockey stick is freely suspended from O and hangs in equilibrium.

Given that the centre of mass of the hockey stick is a distance $\frac{\pi r}{(14 + \pi)}$ from OA ,

- (b) find, in degrees to 3 significant figures, the angle between OA and the vertical.

(2)

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



$$\text{is } 10 \ln \left(\frac{3 + 2e}{5} \right) \text{ m.} \quad (4)$$



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 9 marks)



- 3.** A thin non-uniform semicircular metal plate has mass M and radius a .

The plate is modelled as a non-uniform semicircular lamina with centre O , radius a and mass M , where the mass per unit area at any point P of the lamina is proportional to the distance OP .

- (a) Show that the constant of proportionality is $\frac{3M}{\pi a^3}$ (5)
- (b) Find, in terms of a , the exact distance of the centre of mass of the plate from O . (4)



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[illegible]

4. A light elastic string of natural length a has modulus of elasticity kmg , where k is a constant. One end of the string is attached to a fixed point O and the other end is attached to a particle of mass m . The particle moves, with the string stretched, in a horizontal circle with constant angular speed ω , with the centre of the circle vertically below O .

(a) Show that, if the string makes a constant angle θ with the vertical,

$$\cos \theta = \frac{kg - a\omega^2}{ka\omega^2} \quad (10)$$

(b) Show that $\omega < \sqrt{\frac{kg}{a}}$ (2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 12 marks)



5. A light inextensible string of length a has one end attached to a fixed point O . The other end of the string is attached to a small stone P of mass m . The stone hangs at rest vertically below O , with the string taut, and is then projected horizontally with speed $\sqrt{\frac{22ag}{5}}$. The stone is modelled as a particle.

Using the model,

- (a) show that the speed of P at the instant when the string first goes slack is $\sqrt{\frac{4ag}{5}}$ (7)
- (b) state the magnitude and direction of the acceleration of P at the instant when the string first goes slack, justifying your answers, (2)
- (c) find the speed of P at the instant when it first passes through the horizontal, through O , after the string goes slack. (3)
- (d) Give a reason why your answer to part (c) is unlikely to be the actual speed of the stone at this instant. (1)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 13 marks)



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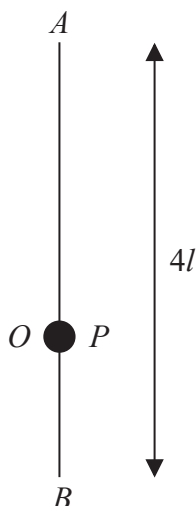


Figure 2

A light elastic string, of natural length l and modulus of elasticity λ , has one end attached to a fixed point A and the other end attached to a particle P of mass m . Another light elastic string, of natural length l and modulus of elasticity 3λ , has one end attached to the particle P and the other end attached to a fixed point B . The point B is vertically below A with $AB = 4l$. The particle P rests in equilibrium, with both strings stretched, at the point O , as shown in Figure 2.

- (a) (i) Show that, if $\lambda > kmg$, where k is a constant

$$AO = \frac{l(10\lambda + mg)}{4\lambda} \quad (4)$$

- (ii) Find the least possible value of k .

(2)

It is now given that $\lambda = 2mg$ and air resistance is modelled as being negligible.

The particle P is projected vertically upwards from O .

Using the model,

- (b) prove that P begins to move with simple harmonic motion of period $2\pi\sqrt{\frac{l}{8g}}$ (6)

- (c) Find, in terms of g and l , the greatest speed with which the particle can be projected vertically upwards from O if P continues to move with simple harmonic motion of period $2\pi\sqrt{\frac{l}{8g}}$ indefinitely. (3)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 15 marks)



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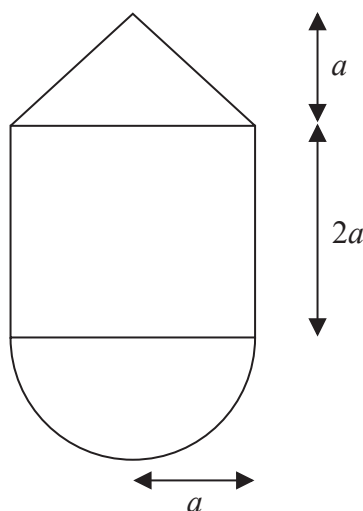


Figure 3

A solid toy consists of a solid hemisphere of radius a , a solid right circular cylinder of radius a and height $2a$, and a solid right circular cone of base radius a and height a , joined together. The plane face of the cone coincides with one of the plane faces of the cylinder and the plane face of the hemisphere coincides with the other plane face of the cylinder. A side elevation of the toy is shown in Figure 3.

The cone and cylinder are made of uniform material of density ρ , and the hemisphere is made of uniform material of density $k\rho$.

The toy is designed so that if it is placed, with its axis vertical and the hemisphere downwards, on a rough horizontal table and then slightly tilted to one side, the toy will return to the vertical position. The table is sufficiently rough to prevent the toy from slipping on the table.

(a) Show, by considering the centre of mass of the toy, that $k > 11$

(7)

If instead $k = 11$ and the toy is placed, with its axis vertical and the hemisphere downwards, on the same rough horizontal table, and then slightly tilted to one side,

(b) describe what happens, explaining your answer.

(3)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 10 marks)

TOTAL FOR PAPER IS 75 MARKS



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Further Mathematics

Advanced

Further Mathematics Option 2

Paper 4: Further Mechanics 2

Sample Assessment Material for first teaching September 2017

Time: 1 hour 30 minutes

Paper Reference

9FM0/4F

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

1. A flag pole is 15 m long.

The flag pole is non-uniform so that, at a distance x metres from its base, the mass per unit length of the flag pole, $m \text{ kg m}^{-1}$ is given by the formula $m = 10\left(1 - \frac{x}{25}\right)$.

The flag pole is modelled as a rod.

- (a) Show that the mass of the flag pole is 105 kg.

(3)

- (b) Find the distance of the centre of mass of the flag pole from its base.

(4)

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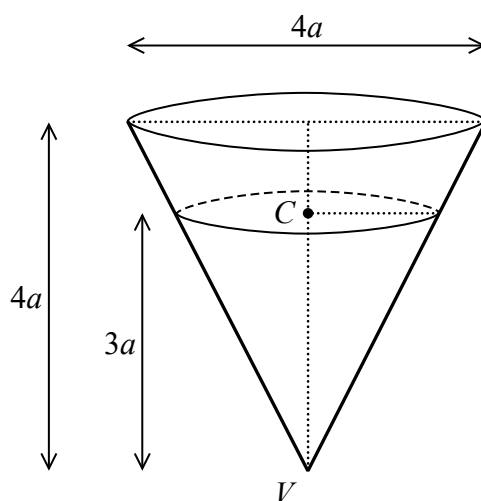


Figure 1

A hollow right circular cone, of base diameter $4a$ and height $4a$ is fixed with its axis vertical and vertex V downwards, as shown in Figure 1.

A particle of mass m moves in a horizontal circle with centre C on the rough inner surface of the cone with constant angular speed ω .

The height of C above V is $3a$.

The coefficient of friction between the particle and the inner surface of the cone is $\frac{1}{4}$.

Find, in terms of a and g , the greatest possible value of ω .

(8)

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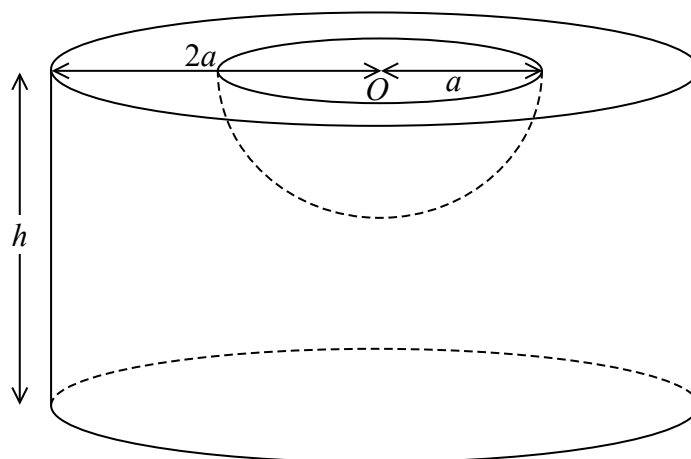


Figure 2

A uniform solid cylinder has radius $2a$ and height h ($h > a$).

A solid hemisphere of radius a is removed from the cylinder to form the vessel V .

The plane face of the hemisphere coincides with the upper plane face of the cylinder.

The centre O of the hemisphere is also the centre of the upper plane face of the cylinder, as shown in Figure 2.

(a) Show that the centre of mass of V is $\frac{3(8h^2 - a^2)}{8(6h - a)}$ from O . (5)

The vessel V is placed on a rough plane which is inclined at an angle ϕ to the horizontal.

The lower plane circular face of V is in contact with the inclined plane.

Given that $h = 5a$, the plane is sufficiently rough to prevent V from slipping and V is on the point of toppling,

(b) find, to three significant figures, the size of the angle ϕ . (4)

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Question 3 continued

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4. A car of mass 500 kg moves along a straight horizontal road.

The engine of the car produces a constant driving force of 1800 N.

The car accelerates from rest from the fixed point O at time $t = 0$ and at time t seconds the car is x metres from O , moving with speed $v \text{ m s}^{-1}$.

When the speed of the car is $v \text{ m s}^{-1}$, the resistance to the motion of the car has magnitude $2v^2 \text{ N}$.

At time T seconds, the car is at the point A , moving with speed 10 m s^{-1} .

(a) Show that $T = \frac{25}{6} \ln 2$ (6)

(b) Show that the distance from O to A is $125 \ln \frac{9}{8} \text{ m}$. (5)

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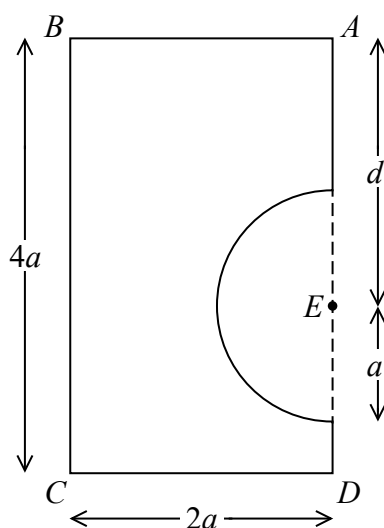


Figure 3

A shop sign is modelled as a uniform rectangular lamina $ABCD$ with a semicircular lamina removed.

The semicircle has radius a , $BC = 4a$ and $CD = 2a$.

The centre of the semicircle is at the point E on AD such that $AE = d$, as shown in Figure 3.

- (a) Show that the centre of mass of the sign is $\frac{44a}{3(16 - \pi)}$ from AD . (4)

The sign is suspended using vertical ropes attached to the sign at A and at B and hangs in equilibrium with AB horizontal.

The weight of the sign is W and the ropes are modelled as light inextensible strings.

- (b) Find, in terms of W and π , the tension in the rope attached at B . (2)

The rope attached at B breaks and the sign hangs freely in equilibrium suspended from A , with AD at an angle α to the downward vertical.

Given that $\tan \alpha = \frac{11}{18}$

- (c) find d in terms of a and π . (6)

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 12 marks)

6. A small bead B of mass m is threaded on a circular hoop.

The hoop has centre O and radius a and is fixed in a vertical plane.

The bead is projected with speed $\sqrt{\frac{7}{2}}ga$ from the lowest point of the hoop.

The hoop is modelled as being smooth.

When the angle between OB and the downward vertical is θ , the speed of B is v .

(a) Show that $v^2 = ga\left(\frac{3}{2} + 2\cos\theta\right)$ (3)

- (b) Find the size of θ at the instant when the contact force between B and the hoop is first zero. (5)

- (c) Give a reason why your answer to part (b) is not likely to be the actual value of θ . (1)

- (d) Find the magnitude and direction of the acceleration of B at the instant when B is first at instantaneous rest. (5)

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Question 6 continued

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Question 6 continued

(Total for Question 6 is 14 marks)

7. Two points A and B are 6 m apart on a smooth horizontal surface.

A light elastic string of natural length 2 m and modulus of elasticity 20 N, has one end attached to the point A .

A second light elastic string of natural length 2 m and modulus of elasticity 50 N, has one end attached to the point B .

A particle P of mass 3.5 kg is attached to the free end of each string.

The particle P is held at the point on AB which is 2 m from B and then released from rest.

In the subsequent motion both strings remain taut.

- (a) Show that P moves with simple harmonic motion about its equilibrium position. (7)
- (b) Find the maximum speed of P . (2)
- (c) Find the length of time within each oscillation for which P is closer to A than to B . (5)

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Question 7 continued

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TOTAL FOR PAPER IS 75 MARKS

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Other names

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Level 3 GCE**

Centre Number

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Candidate Number

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Specimen Paper

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/4C**

**Further Mathematics
Advanced
Paper 4C: Further Mechanics 2**

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1. A particle of mass 0.5 kg is attached to one end of a light elastic string. The other end of the string is attached to a fixed point O on a smooth horizontal surface. The string has natural length 2 m and modulus of elasticity 8 N . The particle moves on the surface in a horizontal circle about O with constant angular speed 1.5 rad s^{-1} .

Find the radius of the circle.

(5)

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Question 1 continued

Handwriting practice lines for Question 1 continued.

(Total for Question 1 is 5 marks)



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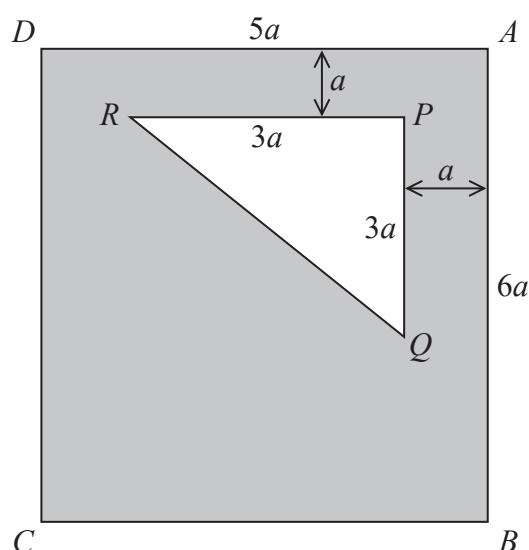


Figure 1

Figure 1 shows a shaded template, T , formed by removing an isosceles right-angled triangle PQR from a uniform rectangular lamina $ABCD$. The rectangle has side AB of length $6a$ and side AD of length $5a$. The triangle has equal sides PQ and PR of length $3a$. Side PQ of the triangle is parallel to side AB of the rectangle and the distance between AB and PQ is a . Side PR of the triangle is parallel to side AD of the rectangle and the distance between PR and AD is a .

- (a) Show that distance of the centre of mass of T from AB is $\frac{44}{17}a$.

(5)

The template T is freely pivoted at A . A horizontal force is applied to T at B so that T rests in equilibrium with AB vertical. The line of action of the force lies in the vertical plane containing T .

Given that the weight of T is 85 N ,

- (b) find the magnitude of the force exerted on T by the pivot at A .

(4)



Question 2 continued

Lined area for writing the answer to Question 2.

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Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 9 marks)



3. A particle, P , of mass 2 kg is attached to one end of a light elastic spring of natural length 0.8 m and modulus of elasticity 49 N . The other end of the spring is attached to a fixed point A . Particle P hangs freely in equilibrium, with P vertically below A .

The particle is now pulled vertically down to the point B , where $AB = 1.4$ m, and released from rest.

- (a) Show that P moves with simple harmonic motion about its equilibrium position. (5)
- (b) Find the length of time, in each complete oscillation, for which P is more than 1.2 m vertically below A . (4)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 9 marks)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 7 marks)



5. For a particle above the surface of the Earth, at a distance x from the centre of the Earth, the magnitude of the gravitational force acting on the particle due to the Earth is inversely proportional to x^2 .

At the surface of the Earth, the acceleration due to gravity is g .

The Earth can be modelled as a fixed sphere of radius R .

A particle P of mass m is at a point that is at a distance $(x - R)$ above the surface of the Earth.

- (a) Show that the magnitude of the Earth's gravitational force acting on P is $\frac{mgR^2}{x^2}$ (3)

A rocket is fired vertically upwards with speed U from a point on the surface of the Earth. When the rocket is at a height $2R$ above the surface of the Earth, the speed of the rocket is $\sqrt{gR}$. The rocket is modelled as a particle of constant mass m and the Earth is modelled as a fixed sphere of radius R . All forces acting on the rocket, other than the Earth's gravitational force, can be ignored.

- (b) Find an expression for U in terms of g and R . (5)

- (c) Suggest one way in which the model could be refined to make it more realistic. (1)



Question 5 continued

Handwriting practice area with 20 horizontal lines.

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Question 5 continued

Lined area for writing the answer to Question 5.

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 9 marks)



6. A particle, P , is moving along the positive x -axis in the direction of x increasing. When the displacement of P from the origin, O , is x metres, the velocity of P is $v \text{ m s}^{-1}$, where

$$v = 9 - \frac{3}{x}, \quad x \geq 1$$

At time $t = 0$ seconds, $x = 1$

At time $t = T$ seconds, $x = 3$

- (a) Find the acceleration of P when $x = 3$

(4)

- (b) Show that $T = \frac{2}{9} + \frac{1}{27} \ln 4$

(6)

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Question 6 continued

Lined area for writing the answer to Question 6.

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Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

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(Total for Question 7 is 12 marks)



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Question 8 continued

Lined area for writing the answer to Question 8.

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Question 8 continued

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(Total for Question 8 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS

